

# Customer Shopping Behavior Analysis

A comprehensive study of transactional patterns, customer segmentation, and product preferences derived from 3,900 retail purchase records.

|                    |                 |                   |                    |
|--------------------|-----------------|-------------------|--------------------|
| 3,900              | 18              | 3.75              | 4                  |
| Total Transactions | Feature Columns | Avg Review Rating | Product Categories |

## 1. Project Overview

Retail businesses generate vast amounts of transactional data with every sale, and unlocking the intelligence buried within that data is what separates reactive companies from proactive ones. This project set out to do exactly that — to transform raw purchase records into clear, actionable insights that decision-makers can act upon with confidence.

Working with a dataset of 3,900 customer transactions spanning multiple product categories, the analysis examines who is buying, what they are buying, how they prefer to shop, and how discount behavior influences spending. The ultimate objective is to equip the business with a nuanced understanding of its customer base so that marketing, pricing, and loyalty strategies can be fine-tuned accordingly.

## 2. Dataset Summary

The dataset comprises 3,900 rows and 18 columns, offering a rich cross-section of customer demographics and behavioral signals. Three broad feature groups structure the data:

| Feature Group         | Variables Included                                                                                          |
|-----------------------|-------------------------------------------------------------------------------------------------------------|
| Customer Demographics | Age, Gender, Location, Subscription Status                                                                  |
| Purchase Details      | Item Purchased, Category, Purchase Amount (USD), Season, Size, Color                                        |
| Shopping Behavior     | Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type |

One data quality issue was identified: the Review Rating column contained 37 missing values, which were addressed during the preprocessing phase described in the next section.

### 3. Exploratory Data Analysis using Python

Before any analysis could begin, the raw dataset required systematic cleaning and enrichment. The entire pipeline was built in Python using the **pandas** library, following a six-step workflow:

|                        |                                                                                                                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Loading           | The dataset was imported using pandas, providing immediate access to <code>df.info()</code> for structural inspection and <code>.describe()</code> for statistical summaries.                                             |
| Missing Data Handling  | Null values in the Review Rating column were imputed using the median rating calculated per product category, preserving category-specific rating distributions rather than applying a global average.                    |
| Column Standardization | All column names were converted to snake_case to ensure consistent referencing across Python scripts and SQL queries.                                                                                                     |
| Feature Engineering    | Two derived columns were introduced: <code>age_group</code> , created by binning customer ages into meaningful cohorts, and <code>purchase_frequency_days</code> , extracted from the existing purchase frequency labels. |
| Data Consistency Check | A correlation check confirmed that <code>discount_applied</code> and <code>promo_code_used</code> were effectively redundant. The latter was dropped to simplify the schema.                                              |
| Database Integration   | The cleaned DataFrame was pushed directly into a PostgreSQL database via a Python connection, enabling seamless SQL-based analysis in the next phase.                                                                     |

#### Descriptive Statistics Snapshot

| Statistic | Customer ID | Age   | Purchase Amount (USD) | Review Rating |
|-----------|-------------|-------|-----------------------|---------------|
| Count     | 3,900       | 3,900 | 3,900                 | 3,863         |
| Mean      | 1,950.5     | 44.07 | \$59.76               | 3.75          |
| Std Dev   | 1,126.0     | 15.21 | \$23.69               | 0.72          |
| Min       | 1           | 18    | \$20.00               | 2.50          |
| Median    | 1,950.5     | 44    | \$60.00               | 3.80          |
| Max       | 3,900       | 70    | \$100.00              | 5.00          |

Table 1 — Summary statistics for key numeric fields

### 4. Data Analysis using SQL (Business Transactions)

With the cleaned data loaded into PostgreSQL, ten structured queries were executed to answer specific business questions. Each query was designed to surface a discrete insight that feeds directly into strategic decision-making.

### Query 1 — Revenue by Gender

Male customers contributed substantially more total revenue (\$157,890) compared to female customers (\$75,191), though this likely reflects a difference in customer volume rather than per-transaction spend, given the near-identical average purchase amounts.

| Gender | Total Revenue (USD) |
|--------|---------------------|
| Female | \$75,191            |
| Male   | \$157,890           |

Table 2 — Revenue split by customer gender

### Query 2 — High-Spending Discount Users

Contrary to the assumption that discounts primarily attract budget-conscious shoppers, the query identified 839 customers who applied discounts yet still spent above the overall average purchase amount. This segment represents an opportunity for premium upsell strategies paired with targeted promotions.

| Metric                      | Value |
|-----------------------------|-------|
| Customers matching criteria | 839   |
| Min purchase in segment     | \$62  |
| Max purchase in segment     | \$100 |

Table 3 — High-spending discount user segment summary

### Query 3 — Top 5 Products by Average Review Rating

Gloves emerged as the highest-rated product with an average score of 3.86, followed closely by Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78). Ratings are tightly clustered, suggesting generally positive customer satisfaction across the catalogue.

| Rank | Product | Avg. Review Rating |
|------|---------|--------------------|
| 1    | Gloves  | 3.86               |
| 2    | Sandals | 3.84               |
| 3    | Boots   | 3.82               |
| 4    | Hat     | 3.80               |
| 5    | Skirt   | 3.78               |

Table 4 — Products ranked by average customer review rating

### Query 4 — Shipping Type Comparison

Express shipping users spent an average of \$60.48 per transaction, slightly higher than the \$58.46 average for Standard shipping customers. While the gap is modest, it suggests that customers who prioritize speed

may also have a marginally higher willingness to pay.

| Shipping Type | Avg. Purchase Amount |
|---------------|----------------------|
| Standard      | \$58.46              |
| Express       | \$60.48              |

Table 5 — Average spend by shipping preference

### Query 5 — Subscribers vs. Non-Subscribers

Non-subscribers account for 2,847 customers and the bulk of total revenue at \$170,436, yet their average spend per transaction (\$59.87) is nearly identical to that of the 1,053 subscribers (\$59.49). This suggests the subscription programme has not yet translated into meaningfully higher per-visit spending — a gap the business can address.

| Subscription | Customers | Avg. Spend | Total Revenue |
|--------------|-----------|------------|---------------|
| Yes          | 1,053     | \$59.49    | \$62,645      |
| No           | 2,847     | \$59.87    | \$170,436     |

Table 6 — Subscriber vs non-subscriber spend and revenue

### Query 6 — Discount-Dependent Products

Five products stand out as heavily reliant on discounts to drive sales. Hat leads with exactly 50% of its purchases made under a discount, followed by Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%). These items may struggle to sustain full-price demand without promotional support.

| Product  | Discount Rate (%) |
|----------|-------------------|
| Hat      | 50.00             |
| Sneakers | 49.66             |
| Coat     | 49.07             |
| Sweater  | 48.17             |
| Pants    | 47.37             |

Table 7 — Products with highest proportion of discounted sales

### Query 7 — Customer Segmentation

Customers were classified into three behavioural tiers based on their purchase history. An overwhelming 3,116 customers qualify as Loyal, affirming a strong retention base. The Returning segment holds 701 customers, while just 83 are classified as New — pointing to a mature customer base where acquisition efforts may need amplification.

| Segment | Number of Customers |
|---------|---------------------|
|---------|---------------------|

|           |       |
|-----------|-------|
| Loyal     | 3,116 |
| Returning | 701   |
| New       | 83    |

Table 8 — Customer segmentation by purchase history depth

### Query 8 — Top 3 Products per Category

Within each of the four product categories, the three most frequently purchased items were identified. Jewelry, Blouse, Sandals, and Jacket claim the top position in Accessories, Clothing, Footwear, and Outerwear respectively, each recording around 160–171 orders.

| Category    | Rank 1           | Rank 2              | Rank 3            |
|-------------|------------------|---------------------|-------------------|
| Accessories | Jewelry<br>(171) | Sunglasses<br>(161) | Belt<br>(161)     |
| Clothing    | Blouse<br>(171)  | Pants<br>(171)      | Shirt<br>(169)    |
| Footwear    | Sandals<br>(160) | Shoes<br>(150)      | Sneakers<br>(145) |
| Outerwear   | Jacket<br>(163)  | Coat<br>(161)       | —                 |

Table 9 — Top 3 purchased items per product category

### Query 9 — Repeat Buyers and Subscription Likelihood

Among customers with more than five previous purchases, 2,518 have not subscribed while 958 have. While repeat buyers do convert to subscriptions at a higher absolute count, the ratio reveals that a significant proportion of high-frequency shoppers remain outside the subscription programme — representing a clear conversion opportunity.

| Subscription Status | Repeat Buyers (>5 purchases) |
|---------------------|------------------------------|
| No                  | 2,518                        |
| Yes                 | 958                          |

Table 10 — Subscription uptake among high-frequency buyers

### Query 10 — Revenue by Age Group

Young Adults are the highest-contributing age group with \$62,143 in total revenue, followed by Middle-aged (\$59,197), Adult (\$55,978), and Senior (\$55,763) customers. The relatively even distribution across cohorts indicates that the product range has broad generational appeal, though Young Adults warrant priority in campaign targeting.

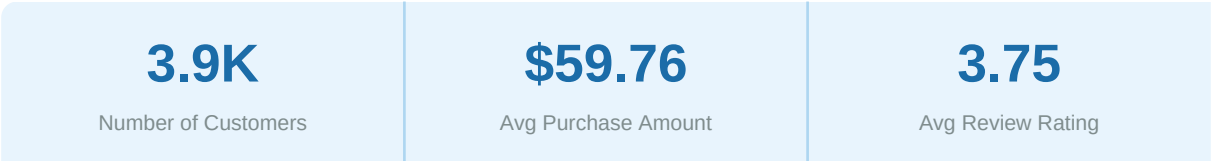
| Age Group | Total Revenue (USD) |
|-----------|---------------------|
|-----------|---------------------|

|             |          |
|-------------|----------|
| Young Adult | \$62,143 |
| Middle-aged | \$59,197 |
| Adult       | \$55,978 |
| Senior      | \$55,763 |

Table 11 — Revenue contribution by customer age group

## 5. Dashboard in Power BI

To make the analytical findings accessible to non-technical stakeholders, an interactive dashboard was developed in Microsoft Power BI. The dashboard consolidates all key metrics into a single view, with slicer controls for Subscription Status, Gender, Category, and Shipping Type allowing users to drill into any customer segment dynamically.



Additional panels in the dashboard visualise subscription status distribution (27% Yes, 73% No), revenue and sales volume broken down by category, and separate bar charts comparing performance across age groups. The layout is designed for executive-level readability while retaining the depth needed for operational insight.

## 6. Business Recommendations

The analysis surfaces five strategic priorities that the business should consider embedding into its near-term roadmap:

|   |                                                                                                                                                                                                                                                                                                      |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Grow the Subscriber Base</b><br>With 73% of customers currently unsubscribed and repeat buyers showing low subscription conversion, there is significant headroom to grow this segment. Exclusive member pricing, early-access launches, or tiered rewards could accelerate uptake.               |
| 2 | <b>Deepen Customer Loyalty Programmes</b><br>The 701 Returning customers represent the most immediate opportunity: targeted incentives designed specifically to push this group over the threshold into the Loyal tier would strengthen lifetime value without the cost of new customer acquisition. |

|   |                                                                                                                                                                                                                                                                                                                                       |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | <p><b>Rationalise the Discount Strategy</b></p> <p>Products like Hat and Sneakers show discount dependency rates approaching 50%. A structured review of margin impact versus volume lift for these items will help determine whether discounts are stimulating genuine demand or simply cannibalising full-price revenue.</p>        |
| 4 | <p><b>Amplify High-Rated Products in Marketing</b></p> <p>Gloves, Sandals, and Boots carry the highest customer satisfaction scores. Featuring these prominently in campaigns — alongside the top-sellers Jewelry, Blouse, and Jacket — aligns promotional spend with proven product strengths.</p>                                   |
| 5 | <p><b>Invest in Young Adult and Express-Shipping Segments</b></p> <p>Young Adults generate the most revenue of any age cohort, and Express shipping users spend slightly more per transaction. Tailoring both creative and channel strategy toward these groups is likely to yield above-average returns on marketing investment.</p> |

*This report was generated from structured analysis of 3,900 customer transactions using Python (pandas), PostgreSQL, and Power BI.*